

SmartEdu – AI-Based Student Performance & Risk Prediction System

Project Overview

A Flutter Web-based Teacher–Student Management System using Machine Learning, Gemini AI, and Face Recognition for student performance monitoring and personalized learning.

Roles:

Teacher

- Manage students & marks
- Upload/assign quizzes and assignments
- Check submissions & give marks
- View student performance and risk

Student

- Manage profile
- View/submit assignments
- Attempt quizzes
- View marks, progress & risk

Machine Learning

Student academic data such as **Quiz Marks, Assignment Marks, Midterm/Exam Marks, Previous Performance and Completion Rate** will be used to predict:

● High Risk | ● Medium Risk | ● Low Risk

Tools: Pandas, NumPy, Scikit-learn, XGBoost.

Gemini AI

Gemini will generate **personalized quizzes and assignments** according to:

Risk Level + Weak Topic + Difficulty + Teacher Notes

High → 5/week | Medium → 3/week | Low → 1/week

Generated content will be stored in **MongoDB** for reuse.

Face Recognition

Before assessment:

Camera → Face Verification → Quiz/Assignment Access

Tools: OpenCV + Face Recognition.

Main Features

- Assignment & Quiz Management
- File Upload/Download/Submission
- Student Progress Tracking
- High/Medium/Low Risk Prediction
- AI Personalized Learning
- Face Verification
- Notification Reminders
- Teacher & Student Dashboards

Technology Stack

Frontend: Flutter Web

Backend: Python + FastAPI

Database: MongoDB + GridFS

ML: Pandas, NumPy, Scikit-learn, XGBoost

AI: Gemini API

Face Recognition: OpenCV

Authentication: JWT

Charts: fl_chart

Email: SMTP

Main Flow

**Marks → ML → Risk → Gemini AI → Personalized Quiz/Assignment → Face Verification
→ Student Attempt/Submit → Marks → Updated Performance**